Semantic Information Extraction and Multi-Agent Communication Optimization Based on Generative Pre-Trained Transformer

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***Abstract*—The collaboration among multiple agents demands for efficient communication. However, the observational data in the multi-agent systems are typically voluminous and redundant and pose substantial challenges to the communication system when transmitted directly. To address this issue, this paper introduces a multi-agent communication scheme based on large language model (LLM), referred to as GPT-based semantic information extraction for multi-agent communication (GMAC). This scheme utilizes an LLM to extract semantic information and leverages the generative capabilities to predict subsequent actions, thereby enabling agents to make more informed decisions. The GMAC approach significantly reduces signaling expenditure exchanged among agents by extracting key semantic data via LLM. This method not only simplifies the communication process but also effectively reduces the communication overhead by approximately 53% compared to the baseline methods. Experimental results indicate that GMAC not only improves the convergence speed and accuracy of decision-making but also substantially decreases the signaling expenditure among agents. Consequently, GMAC offers a straightforward and effective method to achieve efficient and economical communication in the multi-agent systems.**

***Index Terms*—Generative AI, multi-agent, reinforcement learning, semantic communication.**

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I. Introduction

With the rapid development of artificial intelligence, human interaction with machines through various information flows has become increasingly frequent [1]. In various real-world applications (e.g., robotics, sensor networks [2], and autonomous driving systems [3]) can be treated as the multi-agent systems. Agents within these systems may exhibit behaviors that are cooperative, competitive, or a combination of both. Since agents typically have access to partial observations of the environment and cannot fully grasp the overall state, this introduces the challenge of partial observability in multi-agent reinforcement learning (MARL). Furthermore, since all agents need to interact with each other, the action and state spaces of the MARL systems inevitably become huge and challenge to handle [4]. To tackle these challenges, communication among agents becomes crucial. Through the sharing of observations, intentions, experiences, or derived attributes, agents can achieve a deeper comprehension of the environment, which in turn facilitates more informed decision-making processes [5].

Recently, the integration of deep learning methods into reinforcement learning has significantly advanced the field of multi-agent deep reinforcement learning (MADRL). Numerous studies have focused on exploring learning tasks with communication capabilities, aiming to address specific domain challenges through the exchange and sharing of information. For instance, Zhang et al. [6] effectively extracted meaningful information by controlling the variance of messages between agents during the training phase, although this approach may waste some bandwidth due to the exchange of hidden information. Kim et al. [7] focused on the limitations of bandwidth and shared media transmission mediums, and selectively allow only some agents to transmit information, thus imposing restrictions on agent communication. Zhang et al. [8] achieved communication succinctness and robustness by using smoothing and action selection regularizers, but this method is significantly influenced by thresholds. Yuan et al. [9] constructed a teammate model for information transmission, but did not consider the limitations of information transmission volume and communication bandwidth. Although extensive works have utilized communication to enhance overall system performance, there remains insufficient attention to the consumption of data volume and bandwidth resources.

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Fortunately, semantic communication (SemCom) [10], [11], [12], as an emerging technology, has shown significant advantages in next-generation communication systems such as 5G-Advanced and 6G. It focuses on intelligent data transmission, extracting meaningful and task-oriented information, and boasts advantages such as reduced bandwidth usage, lower data transmission volumes, and higher throughput [13], [14], [15] qualities that are essential for multi-agent communication. Semantic communication prioritizes the transmission and reconstruction of valuable semantic information [16], making it particularly suitable for machine-to-machine (M2M) communication, which is beneficial for applications such as smart cities and extended reality (XR) [17], [18]. Therefore, we consider integrating semantic information into multi-agent communication.

Recent studies have shown that generative artificial intelligence (GenAI) can significantly enhance the effectiveness of communication at the semantic layer [19], [20]. GenAI models, such as diffusion [21], flow-based [22], and generative adversarial network [23] models, play a crucial role in compressing semantic information. For instance, Zhang et al. [24] explored the application of GenAI technologies in vehicular networks and proposed a multimodal semantic-aware framework to improve service quality. Zhang et al. [25] explored the use of multi modal large language models to address the out of-distribution issue in image semantic communication. Liu et al. [26] proposed an integrated concept model of AI-Generated Content (AIGC) and SemCom, introducing a content generation layer above the semantic layer, clearly depicting the interaction between AIGC and SemCom. Additionally, Xia et al. [10] proposed a novel GenAI-assisted SemCom network (GenAI-SCN), which integrates global and local GenAI models to provide multimodal semantic content, semantic joint source-channel coding, and AIGC, thereby enhancing resource utilization efficiency and system reliability. Cheng et al. [27] studied a SemCom-empowered AIGC (SemAIGC) generation and transmission framework, achieving efficient content generation through semantic encoders and decoders, and intelligently adjusting computing resources through a resource-aware workload trade-off scheme to optimize service quality and computational efficiency in wireless networks. Tong et al. [28] trained a double Dutch auction auctioneer using a generative pre-trained transformer based deep reinforcement learning algorithm. Yang et al. [29] proposed a novel reinforcement learning-based intelligent agent-driven generative semantic communication (A-GSC) framework, which combines task source information and contextual information, and introduces GenAI to independently design semantic encoders and decoders, effectively enhancing data processing efficiency and accuracy in wireless networks, showing significant performance gains in energy saving and reconstruction accuracy. These studies primarily focus on image content generation, with less research on textual semantic information.

Inspired by the literature [30], this paper conducts an in-depth study on the issue of massive data transmission in multi-agent communication. We find that while the information agents receive frequently varies, the data in successive observations often shows considerable similarity, resulting in significant redundancy in the messages transmitted. We consider these dynamically changing pieces of information as semantic information because they contain key content in the decision-making process, while other stable and unchanging information can be regarded as background information. For example, in autonomous driving scenarios, the vehicle's front camera may capture the same roadside trees for an extended period. Although these observations are continuously changing, their actual impact on navigation decisions is minimal. Therefore, we propose a strategy to transmit messages only when capturing key semantic information, significantly reducing the use of communication bandwidth. Considering the bandwidth limitations in communication environments, this paper proposes the use of LLM to extract key semantic information and perform extreme compression within the semantic domain, forming an efficient semantic chain for transmission. This method not only optimizes the data transmission process but also enhances the efficiency and accuracy of information delivery. The main contributions of this paper are summarized as follows.

- We propose a data transmission strategy based on semantic information extraction aimed at effectively reducing the volume of data transmission in multi-agent systems. In multi-agent communication, we utilize LLM to achieve semantic alignment between observed states and natural language, and compress semantic information. This approach optimizes bandwidth utilization by extracting and compressing information. Additionally, we experimentally confirm the efficacy of the proposed method in reducing data transmission and enhancing transmission efficiency.
- Control decisions in this study are based on predictive outcomes rather than solely relying on the current observed state. We use LLM to predict possible actions in the next moment within the observation and state spaces, thereby prompting agents to take more reasonable actions.
- This paper introduces LLM to establish a bridge between humans and machines, which not only provides a benchmark for subsequent human-machine interactions but may also offer a new interface for humans to access the entire network resource.

Furthermore, the organization of this paper is as follows. Section II provides a concise overview of GenAI technologies in multi-agent communication. Section III elaborates on our methodology. Section IV presents and discusses experimental analysis and performance evaluation results. Finally, Section V summarizes the research findings and outlines future research directions.

II. GenAI-Supported Multi-Agent Communication

This section introduces various GenAI technologies, discussing their advantages and disadvantages, and analyzes their applications in multi-agent communication. Ultimately, we will explore the challenges faced in implementing GenAI

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TABLE I
COMPARISON OF DIFFERENT GENERATIVE MODELS

Technologies	Introduction	Advantages	Disadvantages
Large Language Model (LLM)	Trained on extensive datasets and complex network architectures, the objective is to learn various language patterns and structures.	-Robust Performance -Generalization -Multi-task Learning	-High Computational Resources -Extended Training Duration -Poor Explainability
Generative Adversarial Network (GAN)	A GAN comprises two main components: a generator model and a discriminator model.	-High-Quality Generation -Data Augmentation -Unsupervised Learning	-Training Stability -Resource Intensive -Lack of Control -Evaluation Difficulty
Variational Autoencoder (VAE)	A VAE encodes input data into a latent space and subsequently reconstruct the data from this latent space using a decoder.	-Strong Generation Capabilities -Continuous Latent Space -Unsupervised Learning	-Quality of Generation -Training Challenges -Coverage of Latent Space
Generative Diffusion Model (GDM)	GDM is a type of deep generative model that generate data by simulating the diffusion process from data to noise and its reverse process.	-Good Stability -High-Quality Generation -Wide Applicability	-High Computational Cost -Slow Speed -High Training Resource
Normalizing Flow (NF)	NF transforms a simple distribution into a complex one through a series of invertible transformations, allowing for exact likelihood estimation.	-Exact Likelihood Calculations -Bijective Mappings -Modeling Complex Distributions	-Complexity of Invertible Transformations -Computational Cost -Limited by the Base Distribution

within multi-agent communication and the main issues addressed in this paper.

A. GenAI Technology

GenAI functions by deriving insights from extensive training datasets to execute tasks tailored to user requirements. It primarily relies on several key AI technologies, the specific implementations of which, along with their advantages and disadvantages, are presented in Table I.

B. Applications

As illustrated in Fig. 1, by applying GenAI technologies, multi-agent communication networks can reduce the volume of communication data and decrease the demand for communication bandwidth. The following summarizes some of the main applications of GenAI in multi-agent communication networks.

Variational autoencoders (VAEs) hold significant value for dimensionality reduction and feature extraction of sensor data. The primary reason is their ability to map high-dimensional data to a low-dimensional latent space, achieving data compression, which aids in reducing data processing and communication bandwidth usage. In the latent space, the essential features of the data are preserved, while noise and redundant information are effectively filtered out. In environmental monitoring and intelligent transportation systems, VAEs can analyze data collected from multiple sensors (such as vehicle sensors and roadside units), optimizing traffic flow management and enhancing driving safety. These applications not only improve the efficiency of data processing but also enhance the capabilities of decision support systems, enabling them to more accurately predict and respond to various situations.

Generative adversarial networks (GANs) enhance datasets by generating high-quality synthetic data, which is very helpful for improving the performance of artificial intelligence models in specific tasks. GANs can generate new data that is almost indistinguishable from real data, increasing the diversity of the training set and helping the model learn more generalized features. Although GANs themselves do not directly reduce the use of communication bandwidth and data volume, by locally generating the required data samples, they can reduce the need to download large amounts of data from external sources. This approach is particularly suitable for environments that require large amounts of data but have limited network connectivity.

Generative diffusion models (GDMs), through their unique generative capabilities, can be used to create end-to-end virtual worlds. This capability primarily stems from the model's gradual noise removal process, allowing the generation of highly detailed and realistic images or scenes. Diffusion models can finely control each step of the image change during the generation process, enabling the generated virtual worlds to simulate the dynamics and details of the real world very precisely. Although diffusion models themselves do not directly reduce the use of communication bandwidth and data volume, their ability to generate data can locally produce the required data or scenes, thereby potentially indirectly reducing the demand for high bandwidth and large data transfers.

Large language models (LLMs) can optimize current decisions by predicting future actions [31]. For example, by analyzing historical and real-time data, extracting key information, and predicting future conditions, LLMs can process and analyze vast amounts of textual data. By efficiently extracting semantic information from input data, the volume of data that needs to be transmitted can be significantly reduced. For instance, this method can extract and transmit only key information about changes in traffic flow or accidents, rather than transmitting complete video or image data, thus reducing the demand for high bandwidth and improving data processing efficiency.

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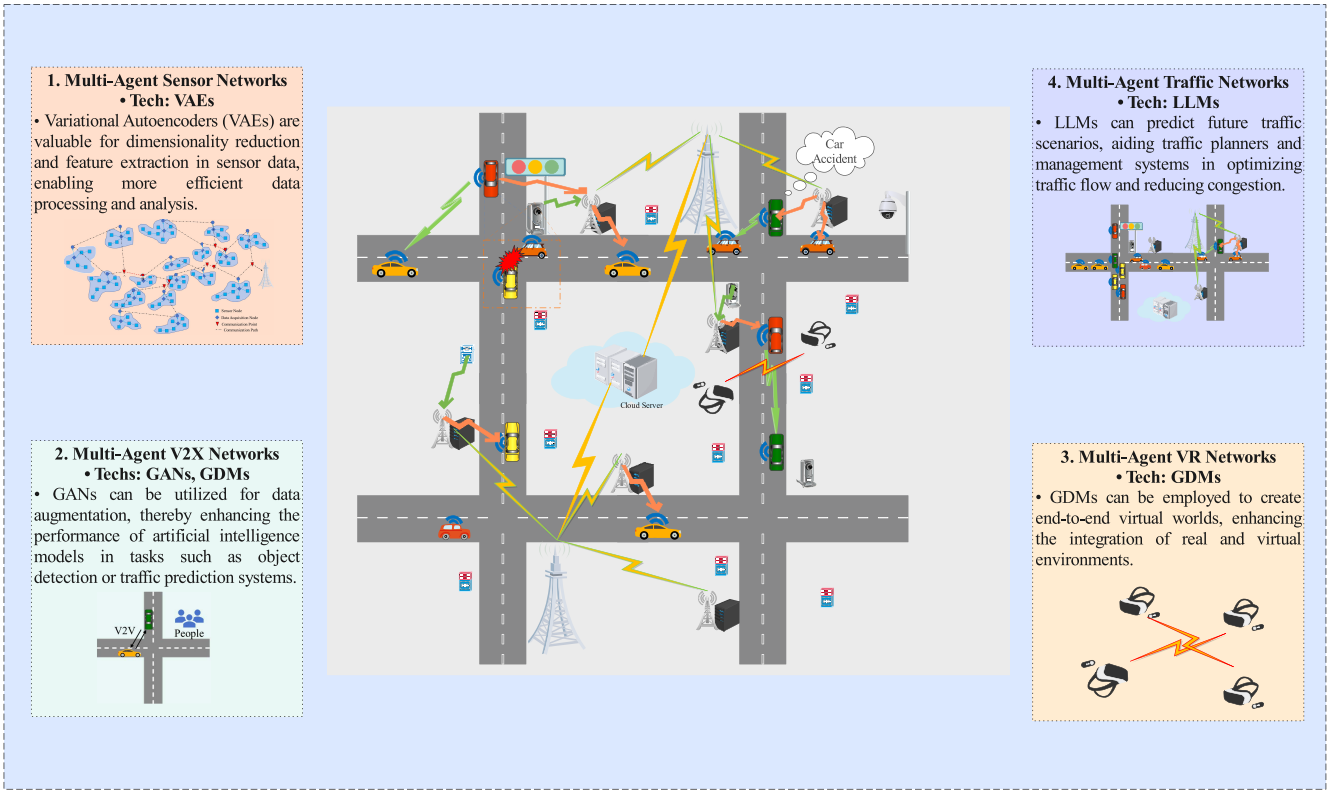

Fig. 1. Applications of GenAI in multi-agent communication networks.

C. Challenges

Despite the significant potential of GenAI in multi-agent communication, it also introduces several challenges, which can primarily be categorized into three areas.

Real-time Data Processing and Prediction: GenAI models need to have the capability to process and analyze large-scale data in real-time and make predictions or decisions. This may involve making multiple requests to specific cloud or edge servers to ensure data reliability. Additionally, delays in data transmission can result in the data not being up-to-date.

Adapting to Rapidly Changing Dynamic Environments: If multi-agent communication networks operate in rapidly changing dynamic environments, GenAI models must be able to quickly adapt and respond to varying environmental conditions. Specifically, the non-real-time nature of data processing can severely impact the quality of communication between agents, thereby negatively affecting subsequent task execution.

Privacy and Security: Addressing data privacy and security issues is crucial in multi-agent communication systems, especially when these systems handle sensitive information. For example, in communication between sensors and in vehicle-to-everything (V2X) communication systems involving vehicles and external devices, data exchange could inadvertently expose user information to potential attackers, thus posing risks of data breaches or privacy violations.

Due to the vast observational results and extensive state spaces in multi-agent communication systems, agents (such as sensors, robots, and smart vehicles) need to continuously transmit these observations, posing significant challenges to the communication system. In our research model, we primarily focus on the communication and decision-making processes among multiple agents. By sharing their respective observational information, agents can gain a more comprehensive understanding of the environment, thereby making wiser decisions. However, due to the substantial redundancy in the observational data, we must effectively handle each agent's partial observation data. To reduce the consumption of uplink communication resources, the sender needs to filter the observations before transmission. This filtering process ensures that only critical information is transmitted, thereby optimizing the data flow and reducing unnecessary network load. This approach not only improves communication efficiency but also enhances the overall response speed and decision quality of the system.

III. GPT-Based Multi-Agent Communication Framework for Decision-Making

In this section, we offer a comprehensive explanation of our proposed GMAC scheme, which is divided into four main stages. 1) information filtering control, 2) semantic extraction and reconstruction, 3) predictive generation, and 4) intelligent decision-making. The overall architecture of the GMAC scheme is illustrated in Fig. 2. Initially, since the local observations of agents are typically not suitable for direct application to LLM, it is necessary to establish a bridge between the natural language domain and the observational state domain. This process, known as semantic alignment, is crucial for ensuring that the LLM can accurately understand the current

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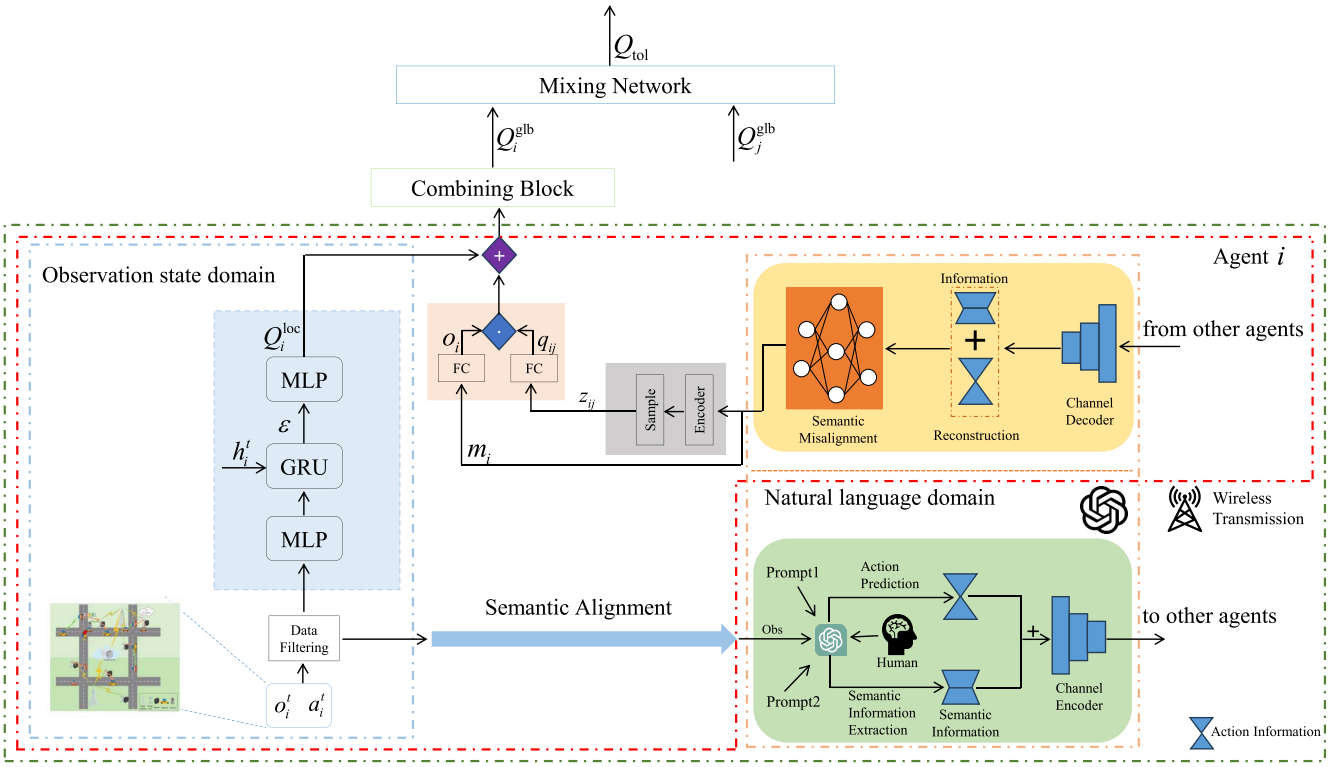

Fig. 2. Overall framework of GMAC.

situation. At the sender side, prompt words are constructed to guide the LLM in extracting and compressing semantic information. These prompt words not only help to construct the scene and task but also guide the LLM in predicting possible actions based on the current observational information. At the receiver side, the received semantic information and predicted action data are reconstructed. Then, these reconstructed pieces of information are used to complete fully cooperative multi-agent tasks. This process enables the multi-agents to make wiser decisions based on more comprehensive and accurate information. Through the collaborative work of these four stages, the GMAC scheme not only optimizes information management but also enhances the decision-making quality of multi-agent systems in complex environments.

A. Information Filtering Control

In multi-agent systems, observational data often exhibit a high degree of redundancy. Direct transmission of all observational data would pose significant challenges to the communication system. Therefore, effective handling of partial observational data for each agent becomes particularly important. The core idea of information filtering control is to reduce the consumption of uplink communication resources by decreasing the number of messages transmitted. To achieve this goal, we define a send message buffer M_i^s for agent i. At time t, agent i calculates the Euclidean distance between the current information It_i and the previous send message buffer M_i^s. Agent i will send a message only if the Euclidean distance exceeds a predefined threshold δ, or the time since the last message was sent exceeds a time threshold t_s. At the receiver, at time step t, newly arrived messages sent by agent i (if any) are first stored in the received message buffer. The validity of each message is then checked. Specifically, the time elapsed since the last update of each message is recorded, and if this period exceeds t_s, the message is considered invalid. These conditions can be expressed with the following formulas.

$$||It_i - M_i^s||_2 \geq \delta, \tag{1}$$

$$t - t_{last_i} > t_s, \tag{2}$$

where t_{last_i} represents the time at which the last message was sent by agent i.

Although the observational values may continuously change, demonstrating the dynamic nature of the environment, the information in consecutive observations often remains similar, leading to a large amount of redundant content in the messages, which has limited value for subsequent tasks. Therefore, filtering observational data not only achieves data compression by eliminating unnecessary or repetitive information, thereby reducing the number of action predictions required at the next time step, but also effectively reduces the burden on the communication system, enhancing overall system performance.

B. Semantic Extraction and Reconstruction

In multi-agent systems, the substantial volume of partial observation data poses significant challenges to communication systems if direct communication is used. This data often contains redundancy, as consecutive observations are similar. We propose that dynamic information, which evolves over time and impacts decision-making, known as semantic

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information, is the most valuable. However, raw observation data are not directly compatible with LLM, requiring effective alignment between observation states and natural language states. This alignment is crucial for LLM to accurately extract semantic information and enhance its understanding of the current situation.

Following semantic alignment, prompts are constructed to utilize LLM as an extractor for extracting semantic information. Generally, after semantic alignment, the information observed from agents can be represented as a discrete set $I = \{I[t_1], I[t_2], \ldots, I[t_i], \ldots, I[t_T]\}$, which includes background information that is irrelevant to subsequent decision-making and semantic information that influences decisions [32]. For example, trees along the sides of roads are generally considered background information because their impact on subsequent navigation decisions is minimal. However, the appearance of obstacles, such as vehicles and pedestrians, constitutes dynamic information, as these elements significantly affect the decision-making process and are crucial components of semantic information.

Therefore, the $I_j[t_i]$ information can be represented as

$$I_j[t_i] = S_j[t_i] + B_j[t_i] \tag{3}$$

where $I_j[t_i]$ represents the information of agent j at the current time step. $B_j[t_i]$ is the background information of agent j at the current time step, and $S_j[t_i]$ denotes the semantic information of agent j at the current time step, where $j = \{1, 2, \ldots, N\}$, denoting the index of agents.

The semantic information emitted by the sender can be expressed as

$$X_j[t_i] = C_t\big(G(I_j[t_i])\big). \tag{4}$$

In these equations, $C_t(\cdot)$ denotes the channel encoder, and $G(\cdot)$ represents the semantic information extractor for the information $I_j[t_i]$. We use a LLM as a semantic information extractor. By constructing prompts, we extract semantic information from the messages, allowing the LLM to select information that holds decision-making value for subsequent tasks.

The receiver receives the semantic information sent by the sender. The received signal can be represented as

$$Y[t_i] = \sum_{k \neq j} (\boldsymbol{H} \cdot X_k[t_i] + \boldsymbol{n}_k), \tag{5}$$

where $\boldsymbol{H}$ represents the channel matrix, and $\boldsymbol{n}_k$ represents the additive white Gaussian noise (AWGN). The decoded signal is represented as

$$X_j^r[t_i] = C_r(Y[t_i]), \tag{6}$$

where $C_r(\cdot)$ represents the channel decoder.

Since the background information has minimal impact on subsequent decisions, it does not need to be transmitted. We only transmit semantic information that has a significant effect on decision-making. We believe that the information received by the recipient includes extracted semantic information and predicted action information $A_j[t_{i+1}]$. The reconstructed information is given by

$$I_j[t_i] = G_r\Big(X_j^r[t_i] \oplus A_j[t_{i+1}]\Big), \tag{7}$$

where $\oplus$ denotes the information reconstruction operation, and $G_r(\cdot)$ represents the semantic realigner, which combines information denoising and semantic realignment tasks.

C. Prediction-Based Generation

The decision-making of intelligent agents is based on predictive results rather than current observations. To achieve this, constructing a large model for prediction prompts is necessary, along with utilizing LLM for prediction, generating semantic chains, and returning them to each intelligent agent. The predictive capability of LLM is closely related to the initial prompt, which is also a standard practice in automated planning. In this paper, we adhere to this principle and input the knowledge base into LLM during the initial phase before collaborative tasks begin. However, constructing a structured knowledge base poses a significant challenge. We found that the optimal combination of a knowledge base needs to be described from three perspectives, including task, rules, and demons [32]. Therefore, this paper will construct prompts from these three perspectives.

##Task
- You are engaged in a specific scenario map in StarCraft II, named XXXX, where your team consists of *x* agents, and the enemy has *x* agents.
- The task is to predict the enemy unit's actions at the next moment based on the information provided by your agents.
...
##Rules
- The output should be a 2 × 2 matrix [[*x*, *x*], [*x*, *x*]], where each row represents one of your agents. Each pair of columns in the matrix represents an action of the enemy unit.
- [Rule 1]: The first column indicates the enemy unit's movement action, where 1 represents moving, 0 represents no movement, and -1 indicates that the unit is not within the visible range.
- [Rule 2]: The second column indicates the enemy unit's attack action, where 1 represents attacking, 0 represents no attack, and -1 also indicates that the unit is not within the visible range.
...
##Demos
- Ensure that the results are returned in the specified format.
- For example, if both of your agents observe the enemy unit moving and attacking, the output should be [[1, 1], [1, 1]].
- If the enemy unit is not within the visible range of any agent, the output should be: [[−1, −1], [−1, −1]].
...

The task component aims to enable LLM to understand the task objectives and information from other collaborating intelligent agents. The rules component specifies LLM's predictions of the enemy's actions in the next time step, indicating which actions can be taken. Within these rules, we can also require LLM to respond using the Chain-of-Thought (CoT) format, making predictions based on this approach rather than directly outputting a prediction plan.

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The demon is an optional component among these three, primarily serving to provide cases for LLM, strengthen LLM's memory, and act according to the specified rules. Generally, the demon should include a description of the scenario task, followed by behavior analysis and action prediction. Through these three components, the LLM can engage in reasoning and generation in subsequent inference stages, ultimately outputting predictions $A_j[t_{i+1}]$ of the enemy's actions for each intelligent agent's actions in the next time step.

D. Multi-Agent Reinforcement Learning Decisions

This paper discusses a multi-agent task characterized by its partial observability, which can be modeled as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP). Defined by the tuple $M = \langle N, S, A, P, \Omega, O, R, \gamma \rangle$, Dec-POMDP involves $N = \{1, 2, \ldots, n\}$ representing the agents, S as the global states, A as the action set, Ω as the observation set. At each time step $t \in T$, each agent $i \in N$ can only acquire the observation $obs_i \in \Omega$ drawn from the observation function $O(s, i)$ where $s \in S$, and then choose the action $a_i \in A$. The joint action $a = \langle a_1, a_2, \ldots, a_n \rangle$ leads to next state $s' \sim P(s' \mid s, a)$ and the global reward $R(s, a)$. Following the execution of the action, the agent receives a reward r_t and transitions to the next state s_{t+1}. The system's objective is to maximize the global value function $Q^{\pi}_{\text{tot}}(\tau, a) = \mathbb{E}_{s,a}[\sum_{t=0}^{\infty} \gamma^t R(s, a) \mid s_0 = s, a_0 = a, \pi]$, where $\gamma \in [0, 1]$ is the discount factor. The goal for the agents is to maximize the joint discounted reward by taking optimal actions. The specific aim is to find a joint strategy that maximizes the global value function. Recent research includes learning a joint Q function $Q_{\text{tot}}(\cdot)$ for cooperative MARL. For instance, in QMIX [34], the joint Q function is represented as a non-negative weighted neural network of $Q_n(o^n_t, h^n_{t-1}, a^n_t)$. Conversely, in the VDN model [33], $Q_{\text{tot}}(\cdot)$ is calculated as the sum of individual Q functions ($Q_{\text{tot}}(o_t, h_{t-1}, a_t) = \sum_n Q_n(o^n_t, h^n_{t-1}, a^n_t)$), where o_t, h_t, and a_t represent the observations, hidden states, and actions of all agents $n \in N$ at time t.

For decision-making, each GMAC intelligent agent inputs its observed values and actions into a gate recurrent unit (GRU) cell to derive a representation of historical information. Subsequently, this historical representation is utilized to compute local q-values, $Q^{\text{loc}}_i(\tau_i, a_i)$, using a multi-layer perceptron (MLP). The observation values of agent j are processed through semantic extraction to generate semantic messages $X_j[t_i] = G(I_j[t_i])$, which are then broadcast to other agents. Upon receiving the messages, agent 1 first decodes and denoises them, then reconstructs the semantic and action information to obtain new messages $I^n_j[t_i] = X^r_j[t_i] \oplus A_j[t_{i+1}]$. The processed information is subsequently fed into the teammate model and message generation network [9], along with local q-values $Q^{\text{loc}}_1(o^1_t, h^1_{t-1}, \ldots)$, to be passed to the combining block. The aggregation block produces global q-values $Q^{\text{glb}}_1(o^1_t, h^1_{t-1}, a)$ for each action a in the action set A. By incorporating messages into the generation of global q-values, better action selection is facilitated as the messages contain global observational information $o_t = \{o^j_t \mid 1 \leq j \leq N\}$ and global historical information $h_t = \{h^j_{t-1} \mid 1 \leq j \leq N\}$ from other agents.

E. Loss Function

The mixed network in the value decomposition method is used to aggregate the global q-values $Q^{\text{glb}}_n(o_t, h_{t-1}, a^n_t)$ from each agent n's network to generate the team's q-values $Q^{\text{team}}_n(o_t, h_{t-1}, a_t)$. The loss function is represented as the expected TD loss and can be expressed as

$$LTD(\theta) = \mathbb{E}_{(\tau,a,r,\tau')\sim\mathcal{D}}\Big[(y - Q_{\text{tot}}(\tau, a; \theta))^2\Big], \tag{8}$$

where $y = r + \max_{a'} Q_{\text{tot}}(\tau', a'; \theta^-)$ is the target network, and θ^- represents the periodically updated target network parameters.

To better integrate information from agents, we apply an attention-like mechanism [37] to compute the query o_i from the agent i's observation O_i, and derive the key $q_{i,j}$ from the representation of teammate j. Queries o_i and $q_{i,j}$ are computed through a simple linear function, and communication weights $\alpha_{i,j}$ are calculated using an exponential function [7]. The communication weight $\alpha_{i,j}$ is then normalized and defines the communication weight between agent i and agent j as

$$\alpha_{ij} = \frac{1}{\eta} \exp\Big(\lambda o_i^T q_{i,j}\Big), \tag{9}$$

where $\eta = \sum_{m \neq i} \exp(\lambda o_i^T q_{i,m})$, and λ is a positive real number serving as a temperature parameter that scales the magnitude of input quantities.

To learn sparse yet effective communication [7], this part of the loss function includes sparse regularization, expressed as

$$L_c = -\sum_{i \neq j}^{n} \alpha_{i,j} \log \alpha_{i,j}. \tag{10}$$

If the current message closely resembles previously sent messages, indicating a lack of semantic information extraction, using past information in q-values calculation may lead to incorrect action selection [6]. To mitigate the issue of message similarity, a regularizer is introduced to maximize the confidence in action selection. This regularizer encourages agents to have confidence in their action choices to avoid being influenced by minor changes in messages. We define a regularizer that maximizes the confidence in action selection as follows

$$L_{\text{tr}} = \sum_{n=1}^{N} \Big(q^{1,t}_n - q^{2,t}_n\Big)^2, \tag{11}$$

where $q^{1,t}_n$ and $q^{2,t}_n$ represent the maximum and second-largest q-values in $Q^{\text{glb}}_n(o_t, h_{t-1}, \ldots)$.

Therefore, the loss function $L(\theta)$ during training can be expressed as

$$L(\theta) = LTD(\theta) + \lambda_c \sum_{i=1}^{n} L_c - \lambda_r L_{\text{tr}}, \tag{12}$$

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Algorithm 1 GMAC

Require: Local observation information of each agent.

for episode $t = 1$ to T **do**
  Control and filter the input information.
  Filter the observation information according to equations (1) and (2).
  Calculate q-values of the filtered information using MLP and GRU.
  Perform semantic alignment on the filtered information.
  Input the filtered information into the large model to predict the next action.
  Extract semantic information from the large model according to equation (3).
  Perform channel encoding for transmission.
  Reconstruct information at the receiver based on semantic and background information.
  Perform semantic misalignment on the reconstructed information.
  Pass the decoded information to the agent.
  Store the sampled network in the value function.
  Multiply the obtained value with the q-values.
  Obtain the global q-values through a combinatorial mixing network.
  Calculate the loss function according to equations (10) and (11).
  Update the target network.
end for

where λ_r is the weight of the regularizer, and λ_c is the adjustable hyperparameter for sparse regularization. The target network is periodically updated by copying parameters from θ, and the replay buffer is consistently refreshed by executing each agent's network and selecting actions using a greedy strategy. The overall GMAC algorithm is detailed in Algorithm 1.

IV. Simulation and Analysis

We primarily focus on the communication and decision-making processes among multiple agents. For this purpose, in this section, we assess the performance of GMAC in multi-agent environments of StarCraft II and Level-Based Foraging (LBF). We assess the performance of GMAC, with a particular emphasis on game win rates (rewards) and communication overhead. We compare GMAC against various baseline methods such as MAIC [9], NDQ [35], QMIX [34], and QPLEX [36], which have demonstrated advanced performance in the StarCraft II Multi-Agent Challenge (SMAC) benchmark tests. MAIC enables each agent to learn to generate incentive messages, thereby achieving effective explicit coordination. NDQ, on the other hand, learns nearly decomposable Q-functions through communication minimization, sending messages to other agents to achieve effective coordination. Initially, we train and configure the in-game AI using the same hyperparameters as those in version 2.4.6 of PyMARL for StarCraft II. The experiments are conducted on a server equipped with two NVIDIA RTX 3090 GPUs, each with 24 GB of memory, and a CPU with 128 GB of RAM. The operating system is Ubuntu 20.04 LTS, and PyTorch 1.13.1 is used. Through these evaluations, our aim is to determine whether GMAC can offer a more efficient and streamlined communication strategy.

A. StarCraft II

1) Winning Rate: We train each algorithm and present the average win rates along with their corresponding 95% confidence intervals for various scenarios in Fig. 3. We particularly focus on the performance in win rates of communication-involved methods such as GMAC, MAIC, and NDQ. Compared to MAIC and NDQ, GMAC demonstrates a faster convergence speed in achieving higher win rates. This advantage may be attributed to GMAC's ability to predict the enemy's next move, providing more informed decision-making guidance for our side, thereby making the decision process more effective. In simple scenarios, GMAC's performance is similar to that of MAIC and QMIX. However, in more complex scenarios (such as 3s_vs_4z, 3s_vs_5z, 2c_vs_64zg), GMAC outperforms MAIC, NDQ, and QMIX. This superior performance is due to GMAC's effective utilization of predicted enemy action information, which accelerates the convergence speed of the training process, thus performing better in more complex environments. In Fig. 3, there are no significant oscillations during the training process, which avoids performance degradation due to instability, indicating that our algorithm is relatively stable. Additionally, the algorithm's winning rate converges across multiple scenarios, demonstrating good convergence properties. These results fully demonstrate the effectiveness and potential of GMAC as a communication strategy in multi-agent tasks, especially in environments that require rapid adaptation and intelligent decision-making.

2) Communication Overhead: To evaluate the information transmission of GMAC, we conduct comparisons with NDQ and MAIC. QMIX and QPLEX are excluded from this assessment as they do not involve communication. To compare the information transmission between agents, the communication overhead is defined as the amount of data transmitted between communicating agents within a total of 200 episodes. Fig. 4 illustrates the information transmission of GMAC, MAIC, and NDQ across different SMAC scenarios, showing that GMAC reduces information transmission by approximately 53% compared to MAIC and NDQ. Given the inferior performance of NDQ relative to MAIC, we focus our comparison between GMAC and MAIC. After reducing the information transmission for both GMAC and MAIC, we evaluate their performance on the 2s_vs_1sc and 3s_vs_4z maps, as shown in Fig. 5. We observe that the win rate of GMAC remains relatively stable with a slight downward trend. This could be attributed to the loss of some information during the semantic extraction process, which leads to a decrease in the win rate. In contrast, as the information transmission is reduced by 50%, the win rate of MAIC drops to around 10%. This indicates that GMAC can effectively extract crucial semantic

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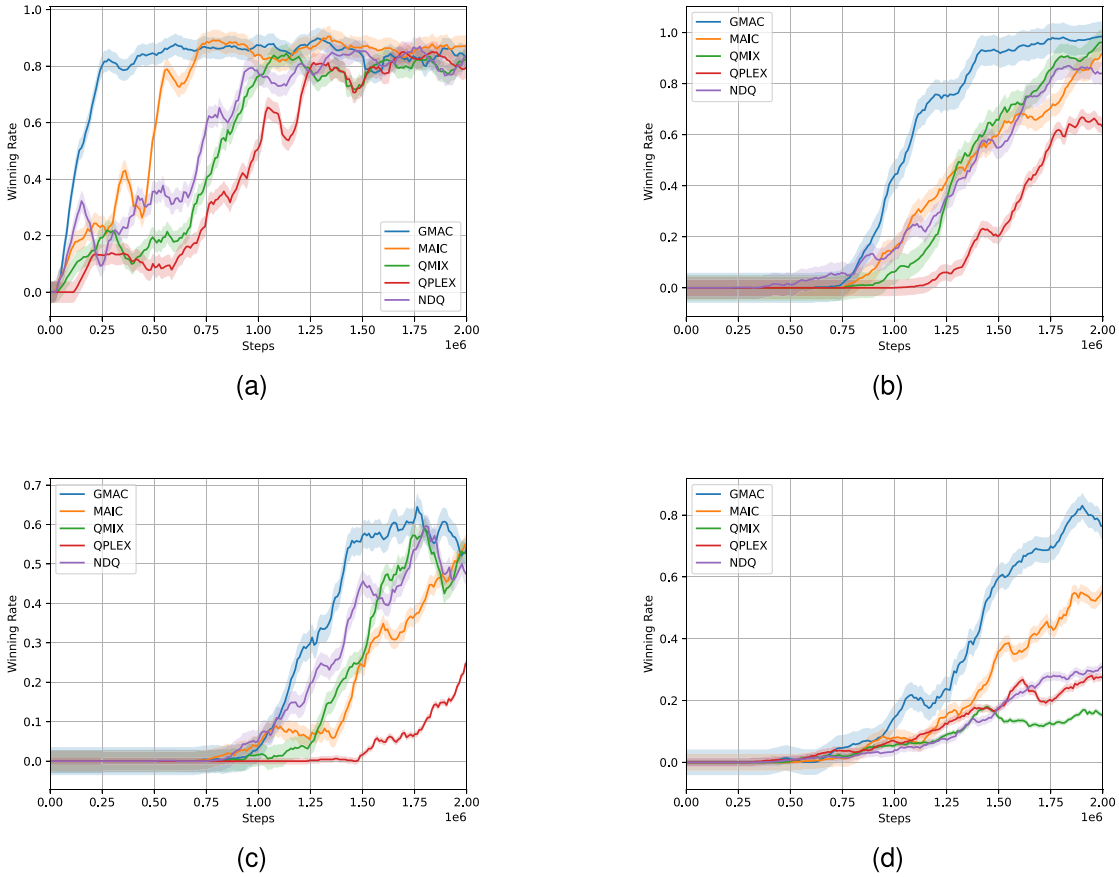

Fig. 3. Winning rates for the four scenarios, shaded regions represent the 95% confidence intervals. (a) 2s_vs_1sc. (b) 3s_vs_4z. (c) 3s_vs_5z. (d) 2c_vs_64zg.

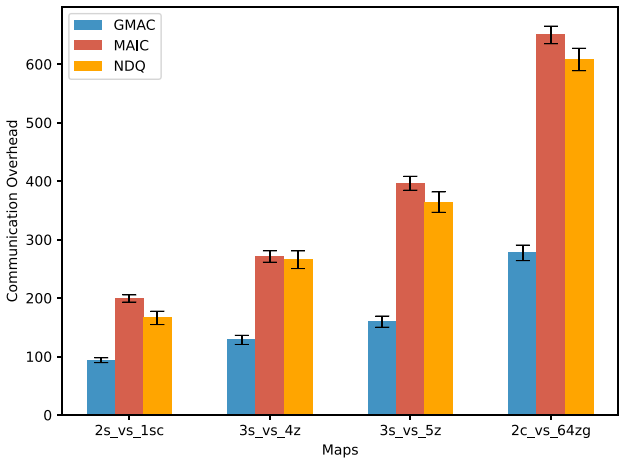

Fig. 4. Communication overhead in GMAC, MAIC, and NDQ across different SMAC maps.

information and significantly reduce communication overhead by eliminating a large amount of redundant information.

To further substantiate the effectiveness of our proposed method, we simulate scenarios of potential information loss during channel transmission. Simulations are conducted in two scenarios, 2s_vs_1sc and 3s_vs_4z, and the average win rates for these scenarios are calculated to assess the performance of different algorithms, as shown in Table II. The more adverse the channel conditions, the more severe the information loss. To ensure a fair comparison, we simulate different message loss rates under the same channel conditions and observe how the win rates and performance of all algorithms vary. The results indicate that when the message loss rate was below 50%, GMAC's performance is superior to other baseline algorithms. NDQ's win rate drops sharply due to a significant reduction in the number of messages, primarily because of changes in the strategy input data. MAIC's win rate also significantly decreases, as it focuses on modeling interactions among teammates without accounting for losses during message transmission. In most cases, the decrease in GMAC's win rate is minimal, mainly due to its ability to effectively extract key information and reduce unnecessary information transmission, allowing it to adapt to harsh channel conditions and thus promote coordination and cooperation among agents.

B. Level-Based Foraging

LBF is a partially observable grid-world game in which agents and food items are initialized with random skill levels [35]. In this game, agents must cooperate to collect food, with the goal of maximizing global gains within a limited field of vision. Specifically, the game is set in an 8×8 grid world where the agent's vision is limited to a range of 2, and the information within this range is used for decision-making. The differences between agents and food necessitate effective cooperation and coordination among agents to achieve higher

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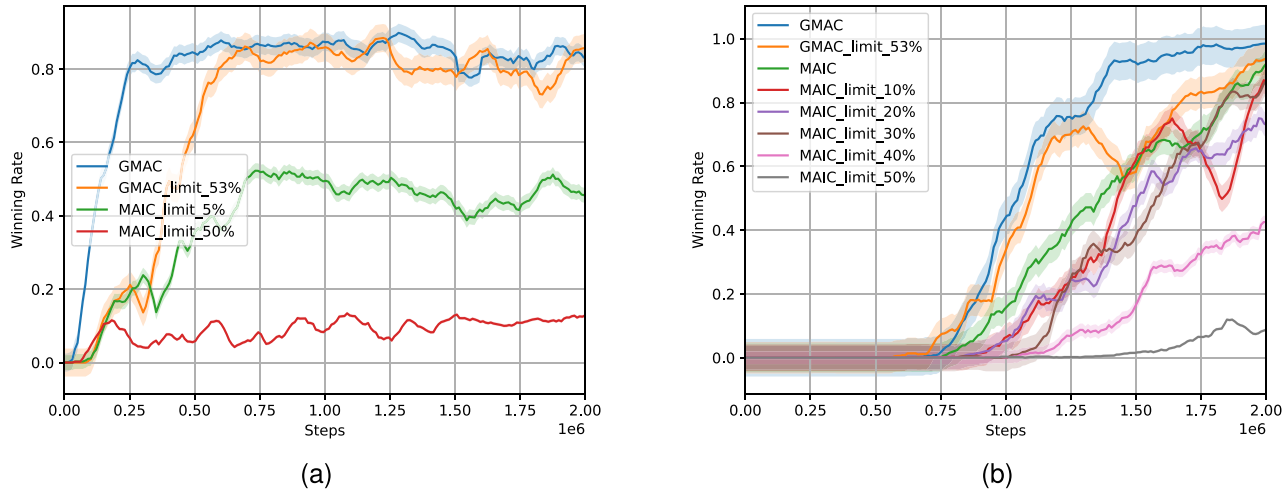

Fig. 5. Communication overhead for the two scenarios and winning rate under different information limits. (a) Winning rate under different limit at 2s_vs_1sc. (b) Winning rate under different limit at 3s_vs_4z.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS MESSAGE PRUNING RATES

Message Lossing Rate	GMAC Test Win Rate %	GMAC Win Rate Increase	MAIC Test Win Rate %	MAIC Win Rate Increase	NDQ Test Win Rate %	NDQ Win Rate Increase
0%	94.6 ± 1.32	0.00%	85.6 ± 3.26	0.00%	81.4 ± 8.37	0.00%
10%	93.8 ± 1.13	-0.81%	71.2 ± 4.43	-14.36%	45.0 ± 11.2	-36.37%
20%	93.2 ± 1.24	-1.35%	62.5 ± 5.41	-23.02%	37.5 ± 9.23	-43.86%
30%	91.5 ± 1.56	-3.12%	65.7 ± 6.51	-19.83%	36.8 ± 5.64	-44.59%
40%	90.4 ± 2.62	-4.18%	34.9 ± 5.38	-50.67%	28.8 ± 12.3	-52.61%
50%	89.7 ± 2.64	-4.89%	19.4 ± 4.26	-64.15%	13.1 ± 8.37	-68.24%

returns. In the LBF game, agents collaborate to gather food. To better guide individual behaviors, the GMAC model is used to predict teammates' actions in the next moment. We compare the GMAC model with multiple baseline methods. These comparisons aim to evaluate the effectiveness and efficiency of GMAC in facilitating agent cooperation.

1) Communication Overhead: To evaluate the performance of GMAC in terms of information transmission, we compare it also with NDQ and MAIC using 200 training episodes. Fig. 6 illustrates the information transmission of GMAC, MAIC, and NDQ in the LBF scenario. The results show that GMAC reduces the amount of information transmitted by over 50% compared to MAIC and NDQ. This significant reduction is primarily because, in the LBF scenario, the transmission involves mainly the location information of agents and foods, which contains a considerable amount of redundancy. Consequently, there is a sharp decline in the volume of useful information that needs to be transmitted. This outcome demonstrates that GMAC can effectively decrease the volume of transmitted data, thereby reducing information redundancy and lessening its impact on the communication system.

2) Reward: In our study, we simulate two scenarios, one in an 8 × 8 grid world where two agents cooperate to collect two food items, and another in a 10 × 10 grid world where three agents collaborate to gather two food items. As shown in Fig. 7(a), the GMAC algorithm achieves an average

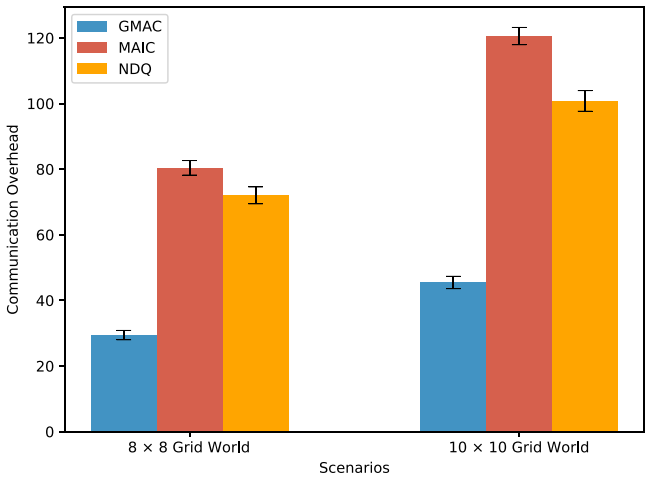

Fig. 6. Communication overhead in GMAC, MAIC, and NDQ across different LBF scenarios.

reward close to 1 in both scenarios, significantly outperforming all other baseline comparisons. This indicates that GMAC effectively utilizes our agents' predicted actions, thereby accelerating the training process and rapidly converging to high rewards. Additionally, an efficient communication structure significantly enhances the coordination among agents. Compared to the QMIX algorithm, MAIC and NDQ demonstrate clear advantages in communication efficiency. This may

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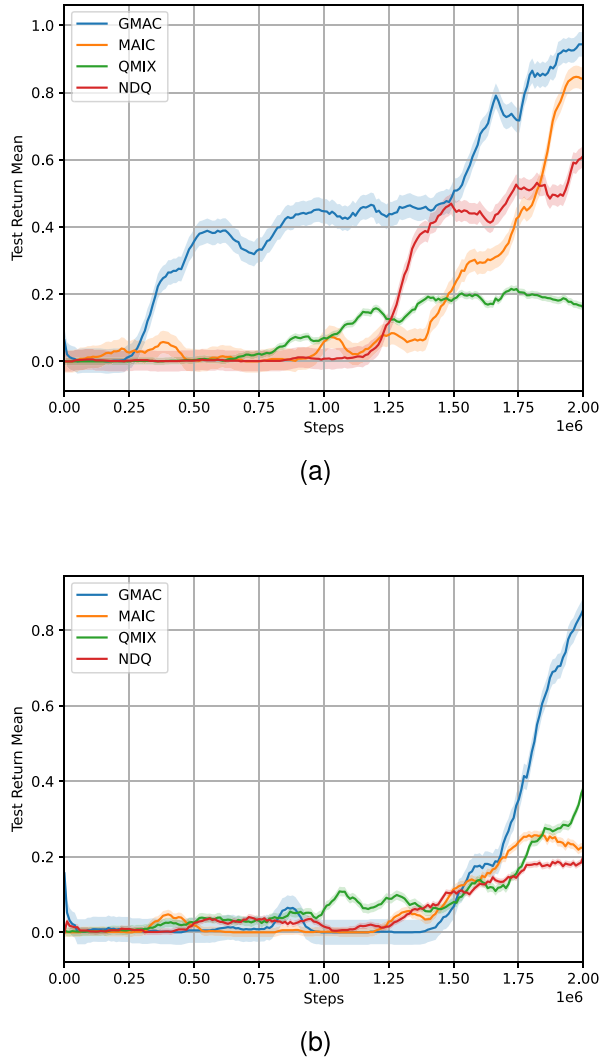

Fig. 7. Reward shaded regions represent the 95% confidence intervals. (a) 8 × 8 grid world with two agents. (b) 10 × 10 grid world with three agents.

be due to the fact that in the LBF environment, two agents needed to effectively communicate and cooperate to collect higher-level food items, thereby earning more rewards. In contrast, the lack of an effective communication mechanism in QMIX likely leads to lower efficiency.

In the more complex 10 × 10 grid environment, as illustrated in Fig. 7(b), although GMAC still outperforms the other three baseline methods, the reward levels for all methods are generally not high. This may be attributed to the competitive relationships among the three agents in this setting. However, GMAC's ability to make more effective decisions using predicted actions indicates that the agents successfully learn these actions, thus maintaining a higher performance level in complex environments.

V. Open Discussion

With the rapid development of LLM and multi-agent communication, this section discusses some future developments and potential challenges in this field.

A. Technical Challenges and Generalizability

Enhancing the scalability of the GMAC scheme to support a larger number of agents and more complex tasks, developing more efficient methods for information filtering control and semantic extraction to improve communication efficiency, improving the robustness of the GMAC scheme in highly dynamic and unpredictable environments, and reducing the computational complexity and resource consumption of the GMAC method while maintaining performance are technical challenges that need to be addressed in future research. These efforts will better integrate LLM with multi-agent semantic communication systems and further optimize multi-agent systems.

B. Privacy and Security Issues

Uploading data to LLM in multi-agent systems may involve sensitive data. For example, in multi-agent communication systems between sensors or V2X communication systems, data exchange may inadvertently expose user information to potential attackers. To address privacy and security issues, the GMAC framework ensures that all data transmitted between agents and LLM is encrypted using advanced encryption standards to protect sensitive information from unauthorized access. Additionally, strict access control mechanisms are implemented to ensure that only authorized agents and systems can access the LLM and the transmitted data.

C. Real-World Challenges and Application Scenarios

Real-world environments are often more complex and unpredictable than simulated ones. Factors such as unforeseen obstacles and varying conditions can pose significant challenges to the robustness and adaptability of the GMAC method. Deploying the GMAC method in real-world applications also involves hardware constraints, which can be addressed by developing edge-side large models. Potential application scenarios for the GMAC scheme include:

- Autonomous vehicle coordination, where multiple vehicles need to communicate and collaborate in complex traffic environments;
- Distributed sensor networks, where sensors need to efficiently share information to monitor and respond to environmental changes;
- Multi-robot systems, where robots need to work together to complete tasks such as search and rescue operations or industrial automation.

In the future, agents will achieve real-time, seamless communication and reasoning through edge-side or on-device large models, thereby improving work efficiency. The integration of large models into agents may be one of the key technologies for achieving general artificial intelligence.

VI. Conclusion

In this study, we introduced GenAI technology to optimize communication among multiple agents by extracting semantic information and predicting subsequent actions. This method has not only accelerated the convergence speed but also significantly reduced the volume of transmitted information. Our

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semantic information transfer method effectively decreased the communication needs between agents. Empirical results have shown that our approach significantly surpasses other baseline methods in various multi-agent collaborative tasks, while reducing communication costs by approximately 53% compared to baseline methods. However, this method has certain limitations, such as weak real-time performance, making it challenging to effectively handle rapidly changing environments.

REFERENCES

[1] H. Wei, W. Ni, W. Xu, W. Jiang, D. Niyato, and P. Zhang, "Semantic communications for multi-view generation," *IEEE Commun. Lett.*, vol. 28, no. 6, pp. 1308–1312, Jun. 2024.

[2] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "Safe multi-agent reinforcement learning for autonomous driving," 2016, *arXiv:1610.03295*.

[3] C. Su, J. Guo, Y. Dong, Z. Chen, V. C. M. Leung, and Z. Han, "Strategy-proof computational resource reservation based on dynamic matching for vehicular edge computing," *IEEE Internet Things J.*, vol. 11, no. 9, pp. 15602–15615, May 2024.

[4] G. Papoudakis, F. Christianos, A. Rahman, and V. Albrecht, "Dealing with non-stationarity in multi-agent deep reinforcement learning," 2019, *arXiv:1906.04737*.

[5] C. Zhu, M. Dastani, and S. Wang, "A survey of multi-agent deep reinforcement learning with communication," *Auton. Agents Multi-Agent Syst.*, vol. 38, no. 1, pp. 1387–2532, 2024.

[6] S. Zhang, Q. Zhang, and J. Liu, "Efficient communication in multi-agent reinforcement learning via variance based control," in *Proc. 33rd Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 3235–3244.

[7] D. Kim, S. Moon, D. Hostallero, and W. Kang, "Learning to schedule communication in multi-agent reinforcement learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019, pp. 1874–1883.

[8] S. Zhang, J. Lin, and Q. Zhang, "Succinct and robust multi-agent communication with temporal message control," in *Proc. 34th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 17271–17282.

[9] L. Yuan, J. Wang, F. Zhang, and C. Wang, "Multi-agent incentive communication via decentralized teammate modeling," in *Proc. 36th AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 9466–9474.

[10] L. Xia, Y. Sun, C. Liang, L. Zhang, and D. Niyato, "GenAI for semantic communication: architecture, challenges, and outlook," 2023, *arXiv:2308.15483*.

[11] E. Hosonuma, T. Yamazaki, T. Miyoshi, A. Taya, Y. Nishiyama, and K. Sezaki, "Image generative semantic communication with multi-modal similarity estimation for resource-limited networks," 2024, *arXiv:2404.11280*.

[12] C. Liang et al., "Generative AI-driven semantic communication networks: Architecture, technologies and applications," 2024, *arXiv:2401.00124*.

[13] X. Kang, B. Song, J. Guo, and Z. Qin, "Task-oriented image transmission for scene classification in unmanned aerial systems," *IEEE Trans. Commun.*, vol. 70, no. 8, pp. 5181–5192, Aug. 2022.

[14] F. Zhao, Y. Sun, L. Feng, L. Zhang, and D. Zhao, "Enhancing reasoning ability in semantic communication through GenAI-assisted knowledge construction," *IEEE Commun. Lett.*, vol. 28, no. 4, pp. 832–836, Apr. 2024.

[15] W. Yang, Z. Xiong, T. Q. S. Quek, and X. Shen, "Streamlined transmission: A semantic-aware XR deployment framework enhanced by Generative AI," 2024, *arXiv:2404.06182*.

[16] Y. Wang, W. Yang, Z. Xiong, Y. Zhao, T. Q. S. Quek, and Z. Han, "Harnessing the power of AI-generated content for semantic communication," 2024, *arXiv:2404.06765*.

[17] S. Tang, Q. Yang, D. Gündüz, and Z. Zhang, "Evolving semantic communication with generative model," 2024, *arXiv:2403.20237*.

[18] H. Du et al., "Generative AI-aided joint training-free secure semantic communications via multi-modal prompts," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2024, pp. 12896–12900.

[19] L. Qiao, M. Boloursaz Mashhadi, Z. Gao, C. H. Foh, P. Xiao, and M. Bennis, "Latency-aware generative semantic communications with pre-trained diffusion models," 2024, *arXiv:2403.17256*.

[20] Y. Liu et al., "Cross-modal generative semantic communications for mobile AIGC: Joint semantic encoding and prompt engineering," 2024, *arXiv:2404.13898*.

[21] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. 35th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 8780–8794.

[22] D. P. Kingma and P. Dhariwal, "Glow: generative flow with invertible 1x1 convolutions," in *Proc. 32nd Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2018, pp. 10236–10245.

[23] I. J. Goodfellow et al., "Generative adversarial nets," in *Proc. 27th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2014, pp. 2672–2680.

[24] R. Zhang, K. Xiong, H. Du, and D. Niyato, "GenAI-enabled vehicular networks: Fundamentals, framework, and case study," 2023, *arXiv:2304.11098*.

[25] F. Zhang, Y. Dong, K. Chen, Y. Shao, and S. C. Liew, "Addressing out-of-distribution challenges in image semantic communication systems with multi-modal large language models," 2024, *arXiv:2407.15335*.

[26] G. Liu, H. Du, D. Niyato, and J. Kang, "Semantic communications for artificial intelligence generated content (AIGC) toward effective content creation," *IEEE Netw.*, vol. 10, no. 9, pp. 295–303, Sep. 2024.

[27] R. Cheng, Y. Sun, D. Niyato, L. Zhang, L. Zhang, and M. A. Imran, "A wireless AI-generated content (AIGC) provisioning framework empowered by semantic communication," 2023, *arXiv:2310.17705*.

[28] Y. Tong et al., "Multi-attribute auction-based resource allocation for twins migration in vehicular metaverses: A GPT-based DRL approach," *IEEE Trans. Cogn. Commun. Netw.*, early access, Jul. 24, 2024, doi: 10.1109/TCCN.2024.3431931.

[29] W. Yang, Z. Xiong, Y. Yuan, W. Jiang, T. Q. S. Quek, and M. Debbah, "Agent-driven generative semantic communication for remote surveillance," 2024, *arXiv:2404.06997*.

[30] A. D. Raha, M. S. Munir, A. Adhikary, and Y. Qiao, "GenAI-driven semantic communication framework for NextG wireless network," 2023, *arXiv:2310.09021*.

[31] P. Wu, Q. Liu, Y. Dong, Z. Wang, and F. Wang, "LMaaS: Exploring pricing strategy of large model as a service for communication," *IEEE Trans. Mobile Comput.*, early access, Jun. 25, 2024, doi: 10.1109/TMC.2024.3418881.

[32] C. Zhang et al., "ProAgent: Building proactive cooperative agents with large language models," 2023, *arXiv:2308.11339*.

[33] P. Sunehag et al., "Value-decomposition networks for cooperative multi-agent learning," in *Proc. 17th Int. Conf. Auton. Agents Multiagent Syst. (AAMAS)*, 2018, PP. 2085–2087.

[34] T. Rashid, M. Samvelyan, C. S. de Witt, G. Farquhar, J. Foerster, and S. Whiteson, "QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 4295–4304.

[35] T. Wang, J. Wang, C. Zheng, and C. Zhang, "Learning nearly decomposable value functions via communication minimization," in *Proc. 8th Int. Conf. Learn. Represent. (ICLR)*, 2020, pp. 1–15.

[36] J. Wang, Z. Ren, T. Liu, and C. Zhang, "QPLEX: Duplex dueling multi-agent Q-learning," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–27.

[37] A. Vaswani et al., "Attention is all you need," in *Proc. 31th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 5998–6008.

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